

Supplementary Material for From Crisis to New Routine: Lasting Shifts in Urban Shopping Mobility and Socioeconomic Inequality post-COVID-19

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1 MOBILITY DATA REPRESENTATIVENESS ASSESSMENT

Table S1 displays the Pearson correlation coefficients (r) between the tracked device counts in the mobility data panel and the population size across various geographic, socioeconomic, and demographic levels. The table also provides additional information, such as the number of observations or groups available for each aggregation. Furthermore, it includes the sum of absolute sampling bias (sum(AB)), median absolute sampling bias (median(AB)), median sampling bias (median(B)), and standard deviation of sampling bias values (sd(B)) for each aggregation, offering further insights into the representativeness of data considering the magnitude and variability of the sampling bias.

The final row of Table S1 presents information about the dataset utilized in the case study of this research. It confirms that the data is reasonably representative, as indicated by the strong correlation between the tracked device counts and the population size in addition to the low level of variation in bias. This suggests that the dataset is reliable and capable of producing meaningful insights about the population under investigation.

Table S1. Basic statistics of mobility data sampling ratio and bias at different geographic, and socioeconomic levels.

Aggregation	Observations	r	sum(AB)	median(AB)	median(B)	sd(B)
Race	7	1.000	2.926	0.1833	-0.0793	0.6486
Household income	16	0.971	7.35	0.4298	-0.1607	0.5627
Educational attainment	24	0.998	5.524	0.1379	-0.0559	0.4112
State	29	0.994	11.297	0.1778	-0.0421	0.6506
County	1,176	0.976	18.108	0.0042	-0.0019	0.0580
CBG	61,437	0.463	45.192	0.0004	-0.0002	0.0025
CBGs preprocessed	5,502	0.756	24.291	0.0035	-0.0029	0.0046

Note: All required data and code for attaining the results shown in this table are available in the repository of this research project: https://osf.io/phbqz/?view_only=0ec27466b42b4d398e959b5d5dec4a71.

2 ANALYSIS OF CHANGE IN DIRECTION OF PREFERENCE SHIFTS

2.0.1 Top 5% socioeconomic communities

In order to gain deeper insights into the direction of temporal preference shifts, we analyze the change in changes of preference parameter values. This change is represented by $\Delta\omega_t$, where ω denotes a preference

parameter and t represents a specific year. The calculation of $\Delta\omega_t$ is illustrated in Equation S1, which provides a quantitative measure for assessing the magnitude and direction of preference changes over a two years period.

$$\Delta\omega_t = (\omega_t - \omega_{t-1}) - (\omega_{t-1} - \omega_{t-2}) \quad (S1)$$

Figures S1 and S2 illustrate the average values of $\Delta\omega_t$ for the top 5% distinctive communities in each of seven single socioeconomic feature, in comparison to the population average. Details regarding the definitions of these distinctive communities can be found in Section ???. $\Delta\omega_t$ values as of 2020 and 2021 separately. Since there are no significant shifts in preference parameters from 2018 to 2019, the $\Delta\omega_t$ values for 2020 are very similar to change in the parameter values from 2019 to 2020, which makes the Figure ?? in the main text and Figure S1 here, convey a similar message. Therefore, in the following, we compare the differences in temporal store selection preferences between communities and the general NYC population from the perspective of each attractiveness factor with a greater focus on values of $\Delta\omega_t$ for 2021, as shown in Figure S2.

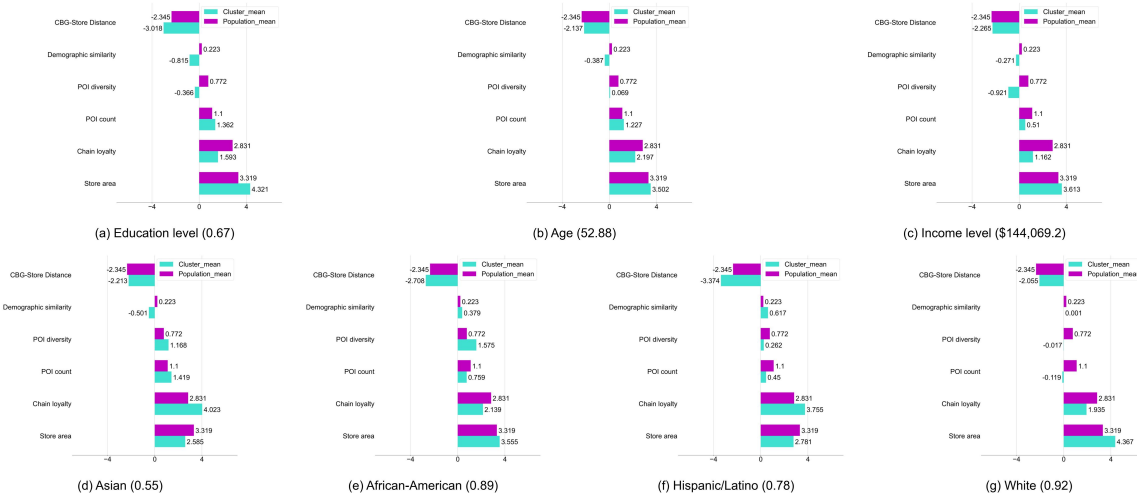


Figure S1. Mean $\Delta\omega_t$ value comparisons in Top 5% socioeconomic communities and population as of 2020. This figure shows the average change in changes of preference parameter values ($\Delta\omega_t$) for the top 5% Socioeconomic Communities in each sub-figure. The community $\Delta\omega_t$ values are shown by blue bars and for comparison purposes the population level $\Delta\omega_t$ values are shown by purple bars. $\Delta\omega_t$ values are computed using Equation S1.

Store area: The community with the top 5% Asian resident proportion showed limited interest in shifting their focus to store area during both 2020 and 2021, indicating strong loyalty to their preferred chains throughout the pandemic and afterward. In contrast, other communities, including those with higher education levels, median incomes, and older age groups, displayed an approximately 20% increased emphasis on store area during the pandemic. This suggests the presence of homogeneous subgroups with socioeconomic features where citizens valued store area more amid the COVID-19 shock. Other communities paid approximately 20% more attention to store area during the pandemic. Interestingly, in 2021, Asian residents seemed indifferent to paying more attention to store area compared to other communities, while communities with higher education levels and median incomes still showed a 20% increase in attention to store area. Additionally, CBGs with residents in the top 5%

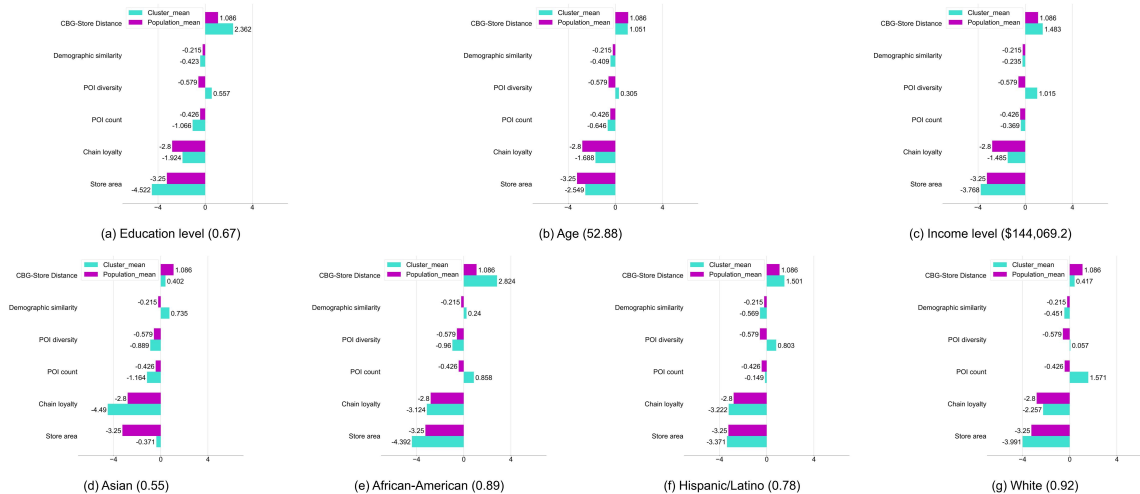


Figure S2. Mean $\Delta\omega_t$ value comparisons in Top 5% socioeconomic communities and population as of 2021.

based on median age exhibited a significant 15% increase in attention to this feature, even greater than the initial increase when the COVID-19 pandemic started to spread. Conversely, African-American residents demonstrated a higher value for store area in their department store selection after the introduction of vaccination and relaxation of social restrictions.

Chain loyalty: The analysis reveals a notable trend among Asian residents, as their preference for chain loyalty experienced a significant increase during and after the pandemic. On the other hand, higher income CBGs displayed a relatively lower level of interest or importance placed on chain loyalty, suggesting a greater degree of indifference towards this factor in their shopping location decisions.

POI count: During the COVID-19 pandemic, residents with higher levels of education, and African-American residents recognized the importance of the abundance of POIs in their decision-making process for selecting department stores. However, in 2021, CBGs with a higher proportion of Asian residents, higher levels of education, and a higher median age among their residents showed a preference for stores that had a greater number of POIs in close proximity. This suggests a shift in the importance placed on the availability and variety of nearby amenities as the pandemic situation evolved.

POI diversity: During the pandemic and as society began to recover in 2021, African-American residents emerged as the leading group in placing a strong emphasis on the diversity and richness of POIs. This indicates their preference for retail locations that offer a wide range of amenities and services to meet their diverse needs and interests.

Demographic similarity: Residents residing in CBGs with higher levels of education and income displayed a stronger inclination towards selecting stores located in neighborhoods that shared greater demographic similarities with their own community. This observation suggests that the decision-making process of residents regarding department store selection may be influenced by the characteristics and preferences of the community they belong to.

CBG-Store distance: As society transitioned into the post-pandemic era, it was observed that CBGs characterized by higher education levels, a larger African-American population, and an older population demonstrated a greater increase in their attention towards this factor compared to the general population.

2.0.2 Clusters by combined socioeconomic characteristics: Five K-means clusters

This section aims to analyze and compare the temporal store selection preferences among five clusters that have been defined using the K-means algorithm and are based on combined socioeconomic characteristics. We also include the general NYC population for comparison. Specifically, we examine the differences in preferences for each attractiveness factor and place a particular emphasis on understanding the direction of change over time. Details regarding the definitions of these distinctive communities can be found in the main manuscript.

Figures S3 and S4 show changes in the value of $\Delta\omega_t$ between 2020 and 2021. It is abundantly clear that from 2020 to 2021, $\Delta\omega_t$ changes reversed roughly among each community and the entire population. People's earlier increased interest in store size, chain loyalty, POI count, and POI diversity from 2019 to 2020 began to decline from 2020 to 2021, while their indifference to CBG-store distance persisted in 2021. The magnitudes of the changes in 2021, nonetheless, are not as large as those in 2020. This occurrence confirms our earlier observation that following the outbreak, the decision patterns of store choice of New Yorkers changed from their pre-epidemic behavior and the impact of COVID-19 has not been fully recovered by 2021.

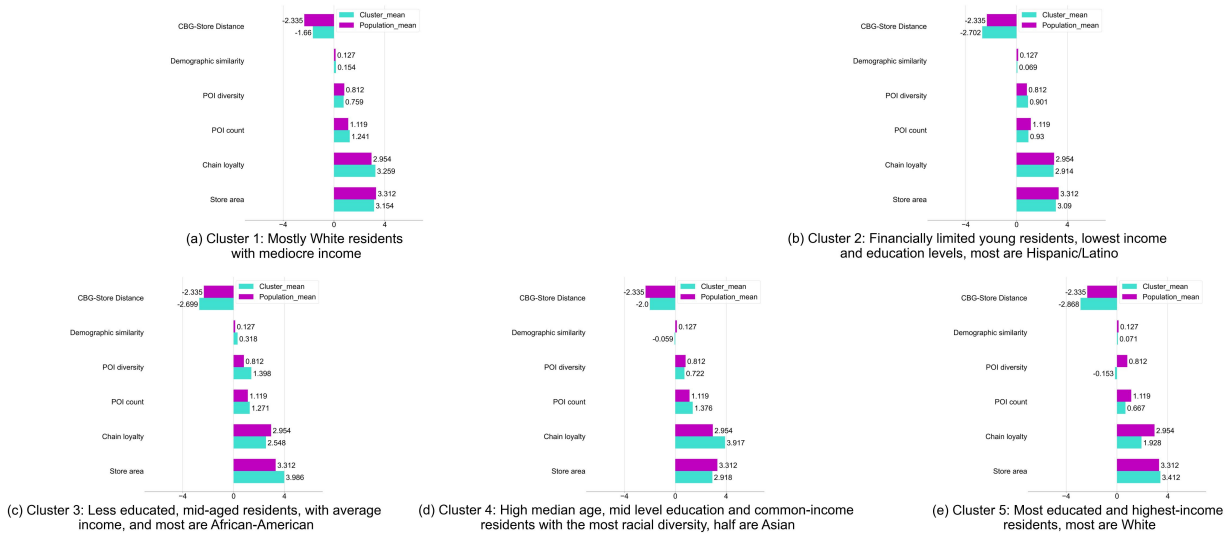


Figure S3. Mean $\Delta\omega_t$ value comparisons in clusters by K-means as of 2020

It has also been observed that changes in $\Delta\omega_t$ due to store area, chain loyalty and CBG-store distance are apparently greater than those in POI count, POI diversity and demographic similarity in 2020, while in 2021 changes in $\Delta\omega_t$ in CBG store distance are similar to those of POI count and diversity. Consequently, it has been signaled that since the outbreak of COVID-19, the products and services that a store can provide have caught more attention instead of the nearby commercial service and the cost of commuting. Overall, our findings using unsupervised clustering confirm that the pandemic response was heterogeneous among different communities.

3 ABBREVIATIONS USED IN THIS PAPER

CBG: Census Block Group

CDF: Cumulative Distribution Function

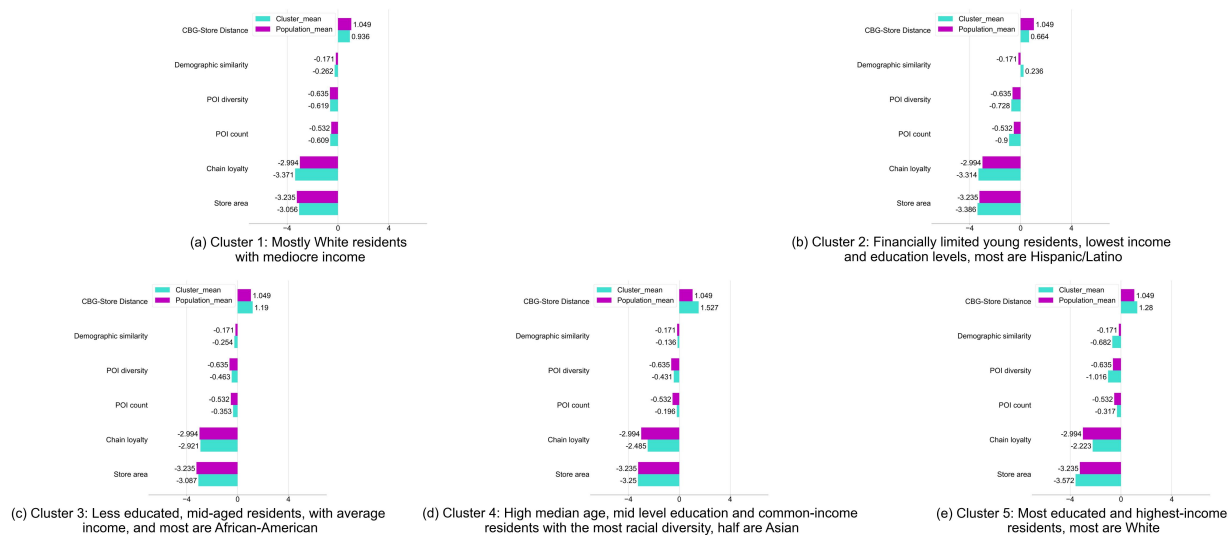


Figure S4. Mean $\Delta\omega_t$ value comparisons in clusters by K-means as of 2021

DID: Difference-in-Differences

GeoDa: Geographical and Spatial Data Analysis Software

GLM: Generalized Linear Model

GWR: Geographically Weighted Regression

MCI: Multiplicative Competitive Interaction

NAICS: North American Industry Classification System

NYC: New York City

OLS: Ordinary Least Squares

POI: Point of Interest

PSO: Particle Swarm Optimization

SA: Simulated Annealing

4 DATA AND CODE AVAILABILITY

The code and pre-processed data used for the analyses in this study are publicly available in the project repository for complete replication and/or further future analysis purposes. The repository is available at: https://osf.io/phbqz/?view_only=0ec27466b42b4d398e959b5d5dec4a71